



fatigue and headaches. A digital signal processor (DSP) is integrated into the solution, which controls the seal or leak of the sound based on sensor fusion; the DynamicVent can take inputs from different sensors, such as a microphone, accelerometer, and barometer, and control the vent to improve the listening experience for the user. The DynamicVent can be employed in TWS, wired earphones, hearing aids, etc.

xMEMS

<https://xmems.com>

PM-MicroLED display driver

SSD2363 from Solomon Systech is the world's first PM-MicroLED display driver for wearables and industrial applications. The SSD2363 improves the display quality and provides a high-brightness display for the intended applications. The display driver not only offers the advantages of traditional display drivers but also provides enhanced durability, better battery life, higher efficiency, and brightness. It can enable designers to build flexible displays that are foldable, rollable, and transparent, which allows more possible display applications in the future. Moreover, the SSD2363 is compatible with a high-resolution PM-MicroLED panel up to 256 RGB x 256. Supporting 768 segments and 256 common outputs, it gives the manufacturers and brands more flexibility in tailor-making 16.7M colour display for panel sizes up to 7.6cm according to their requirements. The driver is suitable for use in a wide variety of applications, such as a display for wearables, home appliances, and industrial applications.

Solomon Systech

<https://www.solomon-systech.com>

Voice-controlled HMI system

R9A06G150 is the first RISC-V MCU from Renesas that is designed for voice-controlled HMI (human-machine interface) systems. The R9A06G150 is a 32-bit ASSP (application specific signal processor) that eliminates the requirement for RISC-V tools and upfront software investment, making it a cost-effective,

production-ready voice control solution. The ASSP supports multiple languages and user-defined keywords for voice-recognition operations.



R9A06G150 is based on a 100MHz CPU with digital

signal processing instructions. It is a specialised audio PDM and SSI interface that offers seamless connection to microphones and codecs. This HMI solution has low power consumption and is compatible with cheap analogue microphones and outputs too. It is suitable for use in applications such as residential and commercial building automation, home appliances, toys, and healthcare devices.

Renesas

<https://www.renesas.com>

LoRa half-duplex transceiver

The LR1121 from Semtech is an ultra-low-power, high-sensitivity LoRa/ (G)FSK half-duplex RF transceiver. It provides multi-band LoRa and long-range frequency hopping spread spectrum (LR-FHSS) communication over sub-GHz and 2.4GHz ISM bands as well as satellite S-band connectivity. LR1121 is designed to comply



with the physical layer requirements of the LoRaWAN specification released by the LoRa Alliance while remaining configurable to meet different application requirements and proprietary protocols. It features an integrated PA regulator supply selector to simplify dual power +15/+22dBm with one board implementation. LR1121 offers support for AES-128 encryption/decryption based algorithms and protects confidential information, such as encryption keys, against unauthorised access. It is suitable for use in a wide variety of applications, such as smart meters, building automation, agricultural sensors, and asset tracking.

Semtech

<https://www.semtech.com>

Software-defined imaging radar

The software-defined imaging radar (SDIR) platform by Zadar combines a powerful radar operating system with novel statistical signal processing algorithms and flexible hardware platforms. It is the first radar processing platform with flexible, software-defined sensors. This platform is an easy-to-use, high-performance sensing solution for intelligent and automated systems that plug directly into a perception stack. Moreover, it improves the potential of radar for a 250x detection performance improvement over existing technology. The SDIR bridges the long-standing gap between sensing hardware and perception software, allowing companies to unlock new levels of performance and safety. It is suitable for use in a wide variety of applications, including automotive, robotics, and smart infrastructure.

Zadarlabs

<https://www.zadarlabs.com>

LoRa-Bluetooth hybrid solution

RAK11720 is a hybrid WisDuo module, which combines LoRa and Bluetooth BLE 5.0 capabilities in a single, compact package. This makes it very flexible and suitable for a wide range of IoT applications. RAK11720 comes



equipped with a powerful AT command set that can be used in tandem with a co-processor or external computer over UART, thus enabling the user to get started right away without writing any code. To simplify the device configuration remotely, the company provides a mobile app, WisToolBox. RAK11720 eliminates the need for a co-processor, thus improving efficiency and reducing power consumption, making the device both compact and lighter. It is suitable for use in a wide variety of IoT applications in smart buildings, factories, or even cities.

RAK Wireless

<https://news.rakwireless.com>